



User Guide (en)

Revision : v.1

AR100

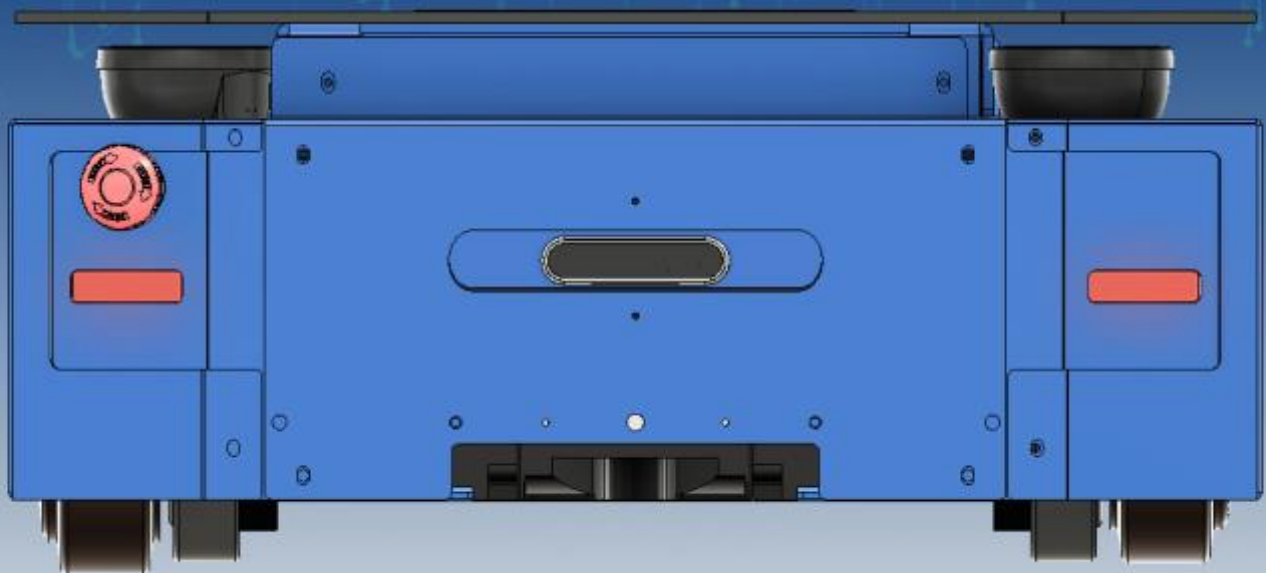


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1. About this document

This product guide demonstrates how the AR100 Autonomous Mobile Robot (AMR) from ANSCER Robotics can be set up and operated. It explains the robot's structure and outlines the maintenance instructions. The guide also provides safety directives and other specifications to securely deploy the robot.

2. Product Introduction

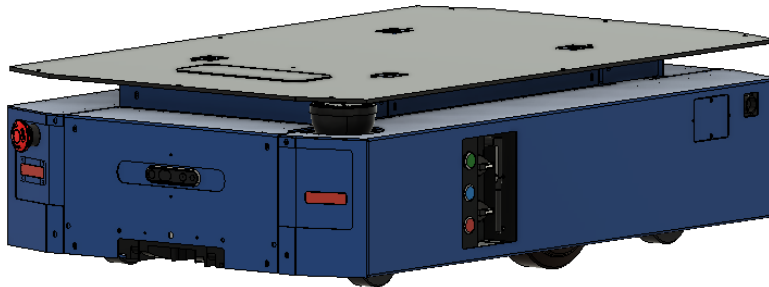


Fig 2.a: AR100

The AR100 is an AMR (autonomous mobile robot) that can lift or tunnel a load of 100 kilograms in warehouses, production facilities, or any dynamic indoor environment. It can detect and avoid unmapped obstacles, including humans.

The AR100 can be programmed and accessed using the ANSCER user interface via desktop computers, laptops, or tablets, using web browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The AMR uses a map of its work environment for navigation. The users can set waypoints and describe missions for the robot through the user interface.

2.1 Main Features

- **Operates in a dynamic work environment.**
The robot can function safely and adeptly in a dynamic ecosystem containing moving obstacles, including human operators and workers.
- **User-friendly**
The user interface has been created to facilitate ease of use. The app can monitor and program the robot without previous experience.
- **Lifts payloads of up to 100 kilograms**
AR100 has been designed to lift, tug, and tunnel loads of up to 100 kilograms in any indoor environment.
- **Convenient and autonomous navigation**
Navigation is autonomous once the map is set up. AR100 is an intelligent robot that can find the most efficient routes to its destination without human interference. It can detect and avoid obstacles, and accordingly adjust its path for the safe transportation of materials across your facility.
- **Signals**
AR100 has both light and sound signals to indicate its status. It has a buzzer that goes off if the robot cannot relocate or find a suitable path for navigation.
- **Automatic slowdown**
The AMR automatically decelerates when it detects obstacles, ensuring the safety of the load it is carrying and the object in front of it.
- **Mapping**
It is possible to manually drive the robot around the facility to map the environment in which the robot operates. During this process, the robot detects and notes all fixed entities in the environment like walls, furniture, and doors. You can later edit this map, or even set other features on it.

2.2 External Components

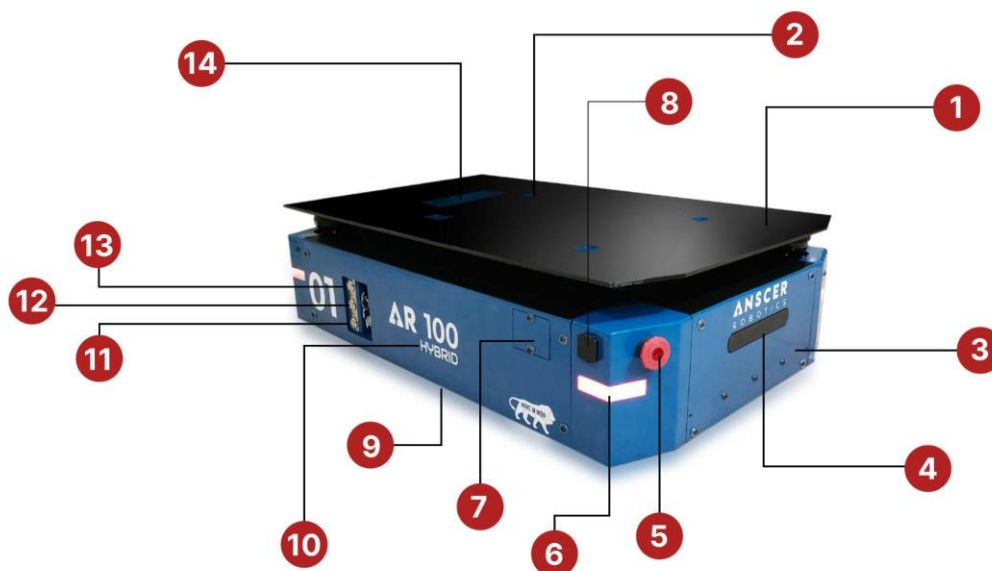
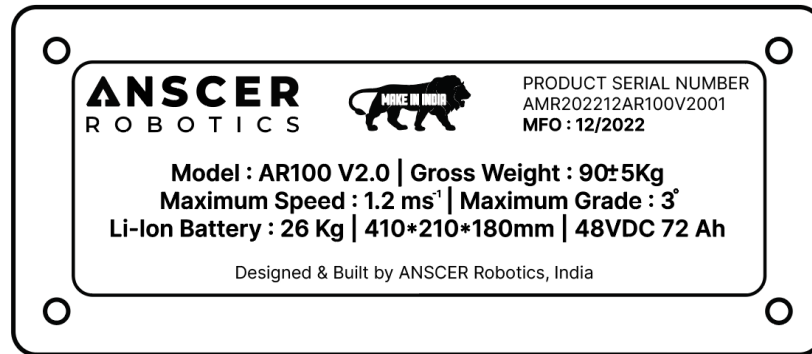


Table 2.2a: Identification of components in Fig 2.2a

Label	Components	Label	Components
1	Top plate	8	Manual charging port
2	Anchor points: 4 pcs	9	Drive wheel: 2 pcs (left and right)
3	Primary hatch: 2 pcs (front and back)	10	Side panel: 2 pcs
4	Camera sensor: 1 pc	11	Reset Button
5	Emergency buttons: 2 pcs (on diagonally opposite corners)	12	Stop button
6	LED indicators: 4 pcs	13	Start Button
7	Debugging Port	14	Auxiliary connectors for external communication

Nameplate

Every AR100 Robot is delivered with a nameplate that is attached to the robot. In addition to the ANSCER logo, the nameplate of AR100 mentions the Product Serial Number, Model Number, Gross Weight, Maximum Speed, Maximum Grade, Battery Weight, and Robot Dimensions.



Control panel

The control panel of the AR100 is on the left side of the robot.

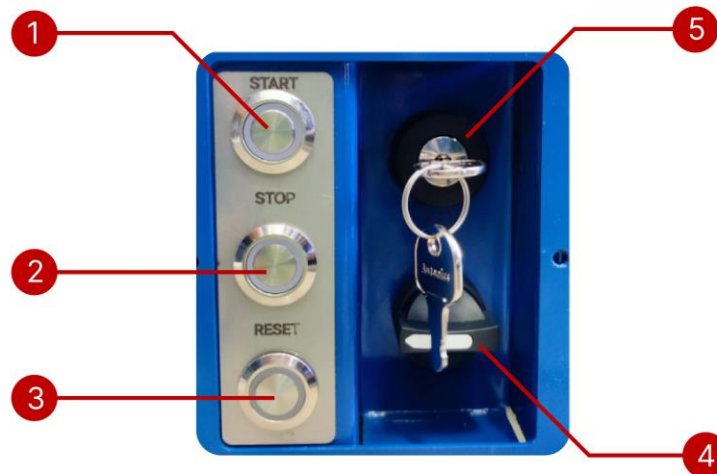


Table 2.3b: Identification of components in Fig 2.3b

Label	Component	Label	Component
1	Start button	4	Auto/ Manual/Brake Release
2	Stop button	5	Key
3	Reset button		

Start Button – Used to turn on the robot.

Stop Button – Used to stop the robot.

Reset Button – Used to reset the robot.

Operating modes:

AR100 has Three modes: **Manual Mode, Autonomous (or Auto) Mode and Brake Release Mode**

Manual Mode

To use this mode, the toggle switch on the robot should be turned to Manual Mode. Here, you can drive the robot manually using the joystick in the robot interface. Only a single person can control the robot manually at a time.

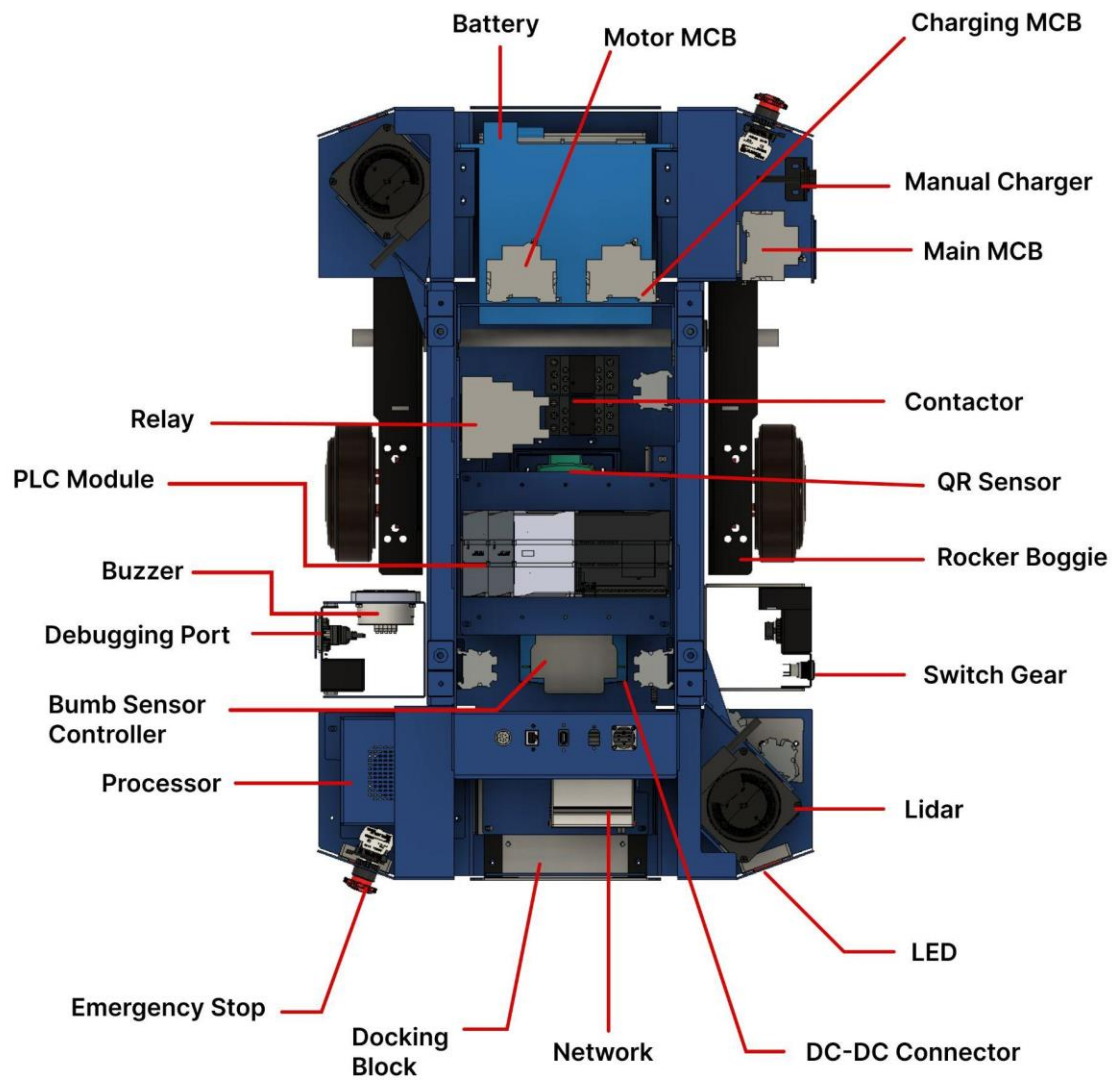
Autonomous Mode

To enable Autonomous Mode, the toggle switch on the robot must be turned to Auto. In this mode, the robot executes the programmed missions autonomously.

Brake Release Mode

The **Brake Release Mode** allows you to push the robot forward and disconnects brake from the Motors and its power source. This mode is useful when robot battery is down, and it is unable to move.

2.3 Internal Components



3. Safety

Please read the safety instructions before setting up and using the robot.

3.1 Key for Safety Messages

- **WARNING:** Indicates fatal or hazardous events that can cause severe damage, death, or injury.
- **CAUTION:** Indicates dangerous events that may cause minor damage or injury. Please follow the instructions carefully to avoid undesirable outcomes.
- **NOTICE:** Indicates important information that must be heeded to avoid issues related to safety and damage of person(s), equipment, and/or property.

3.2 Safety Instructions

Intended Use

The AMR purchased has been designed specifically for its intended use as described by the instructions in this product manual.

Any personal injury, harm, or loss caused to the user by ignoring and violating the instructions provided in this product manual is the user's sole responsibility.

Risk from Misuse

The AMR is to be used only for its intended purpose and design. Ignorance of operation instructions and safety precautions that are provided in this manual, could lead to damage to property, serious personal harm, or even death.

It is recommended that the user follows the given general guidelines:

- Strict adherence to the instructions, roles, and safety guidelines in this manual.
- Use the purchased AMR only for its intended purpose and design.
- Avoid exposing the AMR to any rough weather conditions, rough flooring, explosive air conditions, or any deemed to be dangerous or hazardous conditions.

- Do not modify any part, design, or hardware/software components of the AMR.
- It is recommended that only qualified or trained personnel operate the AMR.

Specific Dangers and Safety Warnings

This section describes certain specific dangers and residual risks that the users are exposed to in the event of operating the AMR.

WARNING

Danger caused by electric current:

- There can be danger caused by the insulation or components which can be dangerous to the user.
- If any damage to the insulation is suspected, switch off the power supply immediately and repair the damaged components.
- Only professional electricians must be allowed to work on the electrical components.
- The system needs to be de-energized for work to be done on the electrical system.
- Switch off the power supply before any maintenance/cleaning/repair tasks.
- Use the correct amperage while changing fuses.
- Do not allow any moisture near the conductive part components, as this could lead to short circuiting.

Danger caused by moving components:

- A user can be injured due to rotating/linearly moving components.
- Do not place any body parts close to the moving components.

Danger caused by malfunctioning safety machinery:

- All safety equipment in the AMR needs to be maintained consistently in good working order.
- Before operating the AMR, check whether the safety equipment of the AMR is in optimum working condition as per standards set in the manual.
- Do not, under any circumstance, disengage the safety equipment.
- Do not modify the safety equipment.

CAUTION

- The AMR should not be used to transport people.
- Avoid inclinations of more than 3° that fall on the AMR's route.
- The AMR must only be on a route that has dry and even surfaces. Uneven surfaces could be hazardous.
- The AMR must not be overloaded beyond its payload limit. The maximum payload for the AR100 is 100 kilograms.

NOTICE

- Ensure that the load for transportation via the AMR is properly and securely mounted on top of the AMR.
- Ensure that there is no leakage of any fluids during transport.
- Ensure that the user is using the charger that is supplied by ANSCER Robotics only.
- Ensure that the AMR is not operated irresponsibly or playfully, as this could be dangerous for the user and those around the user.

Warning Icons on the AMR

Icon	Meaning
	This is a warning icon that warns to be careful of the 48 Volts power supply while using/changing the battery of the AR-1250.
	This is a warning icon to ensure that only an Authorised operator is using/operating the AMR.
	This is a warning icon that reminds the user that the AMR is an automatic vehicle, and hence should be aware of the features and risks involved.
	This is a warning icon that warns the users against riding on the vehicle.
	This is a warning icon that warns against any dangerous electrical situation.
	This is a warning icon that warns against dangerous magnetic fields.
	This is a warning icon that warns against any fire risks.

3.3 Users

The AR100 is only intended to be used by personnel that have received training in their required tasks.

There are three types of intended users for the AR100: Super Admin (Commissioners), Admin (Operators), and direct users.

Commissioners

Commissioners have thorough knowledge of all aspects of commissioning, safety, use, and maintenance of AR100, and have the following main tasks:

- Commissioning the product. This includes creating maps and restricting the user interface for other users and making brake tests with a full payload.
- Conducting the risk assessment in the environment.
- Determining the payload limit, weight distribution, safe fastening methods, safe loading and unloading of loads on AR 100.

Operators

Operators have thorough knowledge of the AR100 and of the safety precautions presented in this user guide. Operators have the following main tasks:

- Servicing and maintaining the AR100.
- Creating and changing missions and map features in the robot interface.

Direct Users

Direct users are familiar with the safety precautions in this user guide and have the following main tasks:

- Assigning missions to AR100.
- Fastening loads to AR100 securely.
- Loading and unloading loads from a paused AMR.

All other persons in the vicinity of AR100 are considered indirect users and must know the suitable conduct/possess awareness of appropriate behavior when they are in the vicinity of the robot.

3.4 LED Indicators

The AMR uses different LED indicators to show the AMR's status of upcoming operation.

The following list gives the LED indications.

Light Indication	
Red Blink	Low Voltage
Red	Error
Yellow Blink	Slow Zone
Yellow	Safety Zone
Green	Mission Success
Blue	Mission Active
Pink	Mission Abort
Cyan Blink	Auto charging
Cyan	Manual Mode
White	Default

4. Get Started

This section describes how to get started with the AR100.

4.1 In the Box



The box will contain:

1. AR100 Robot
2. AR100 User Manual

4.2 Unpacking the Box

To unpack the box, follow the steps below:

Step 1: Place the box in an open area with free space surrounding all sides of the box.



Step 2: Remove the nuts and bolts from the top compartment and take the top lid off.



Step 3: Take the printed documents and accessories out of the box.

Step 4: Remove the nuts and bolts from the front compartment and remove the front wooden slab in a way that it slants down and imitates a sloping rampway.



Once the top and front compartments of wooden slabs are removed, there are two methods of removing the AMR out of the box.

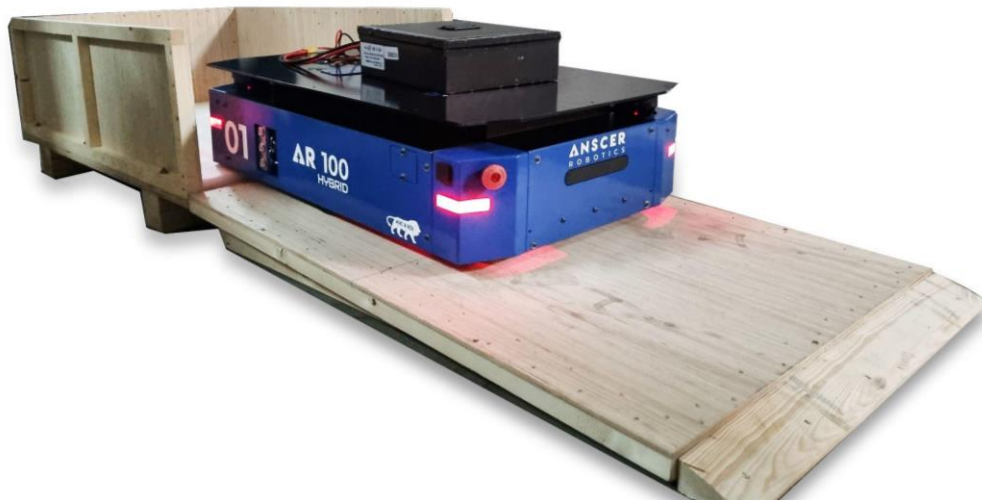
4.3 Taking AR100 out of the Box.

Recommended Method

Step 1: Remove the front compartment so that it acts as a rampway for the AMR to be driven outside.



Step 2: Drive the AMR out of the box using the slanted rampway.



Please note: Drive the robot out carefully to avoid scraping the sides of the box and the AMR.

Alternative Method

This alternative method involves eye bolts and a forklift. Here, we will be taking the AMR out of the box from the top.

Step 1: Remove the top cover.

Step 2: Ensure everything other than the AMR is taken out of the box.

Step 3: Attach the eye bolts securely on the top of the robot.

Step 4: Lift the robot out of the box using a forklift, by attaching heavy-duty lifting straps to the eye bolts.

Step 5: Place the robot slowly and steadily onto a clear area.

Please note: Eye bolts are provided in the box.

CAUTION:

Take precautions while using the forklift and eye bolts. Ensure they are connected properly to avoid accidents.

4.4 Powering ON the AMR

IMPORTANT:

Before powering on your AR100, there is an additional protective measure step that needs to be followed for the AR100:

In the side panel of the AR100, there is a switch called the MCB switch. This switch will be turned off when you receive your packaged AR100. Turn the MCB Switch ON before powering on your AR100.



Steps to Follow

Follow the steps below to power on your AMR.

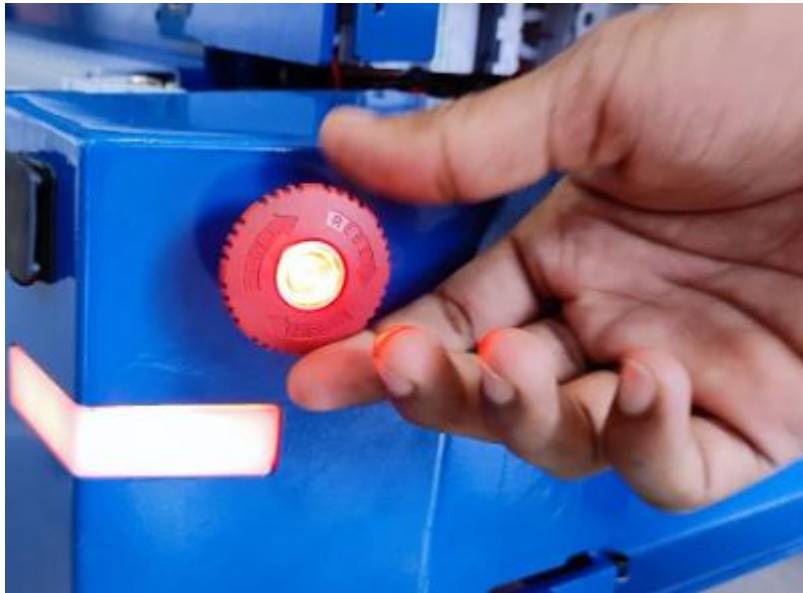
Step 1: Switch on the robot by inserting the provided key and turn it to the right (clockwise).



Step 2: The cyan-blue color of the LEDs indicates that the AMR has begun the software initialization process.



Step 3: Before logging in to the software interface, ensure all the Emergency Protective Stop buttons are released. These red buttons will be pushed inside. To release them, twist the button once.



Step 4: Press the green button to confirm that all the Emergency buttons have been released.

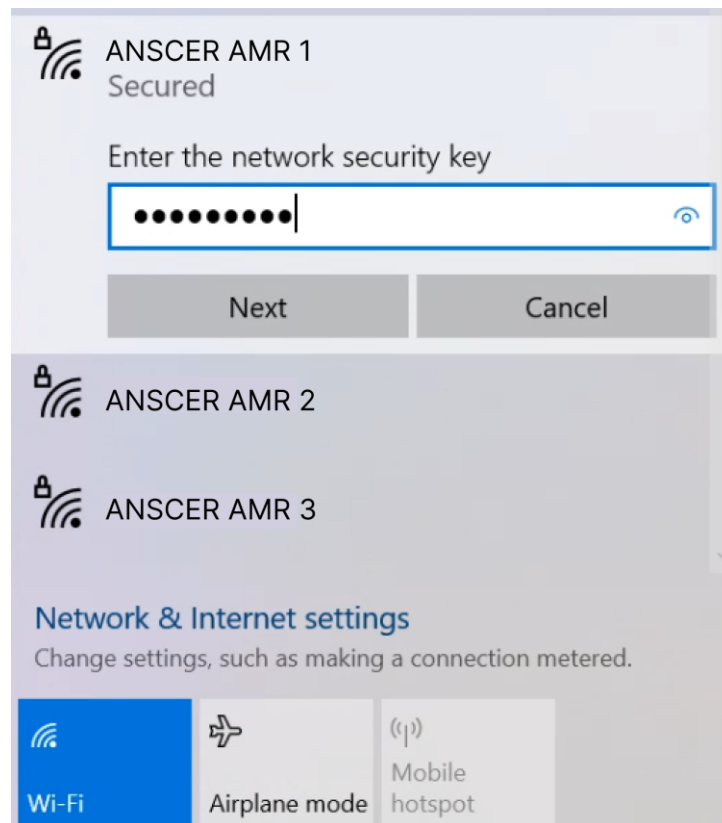


Your AR100 is now ready for operation.

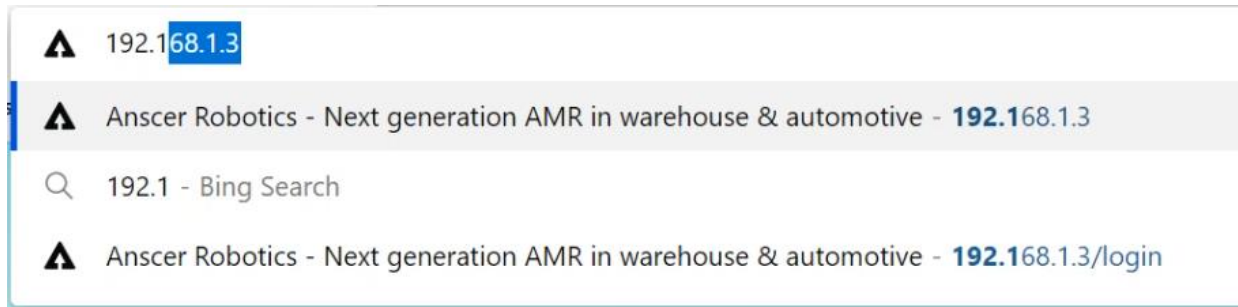
4.5 Connecting and Logging in to the Interface.

Once the AMR is powered on, connect the AMR and your tablet/PC to operate the robot.

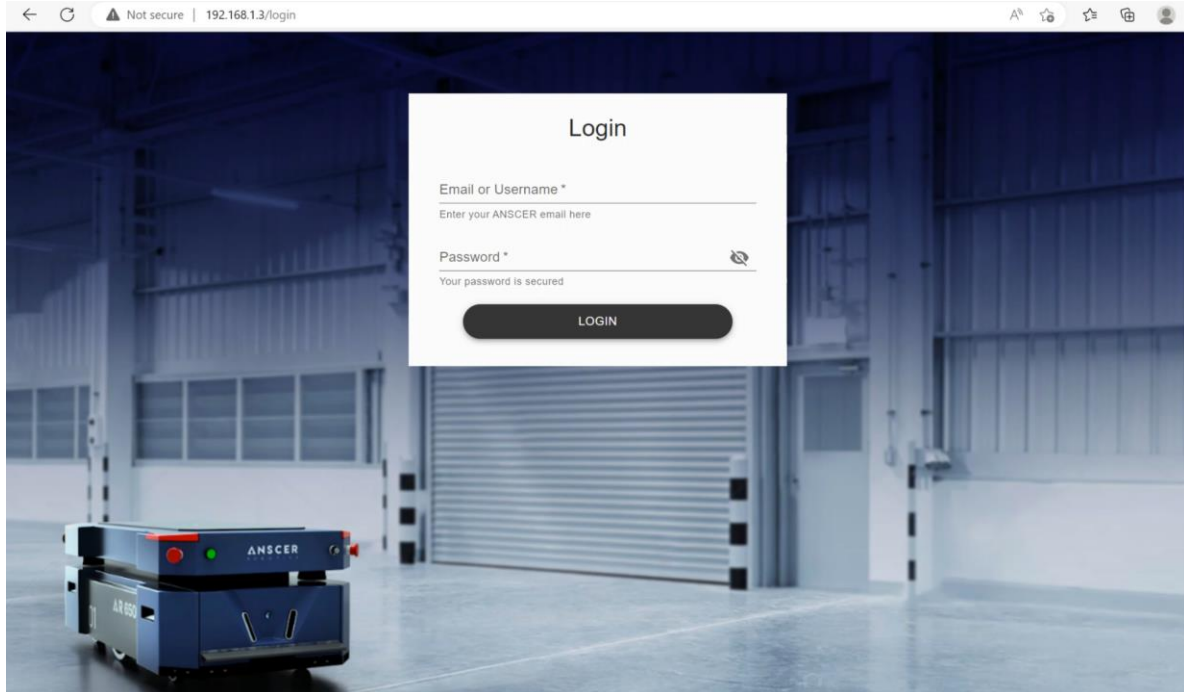
Step 1: Using your PC, tablet, or phone, **connect to the Wi-Fi access point of the robot** using the unique password for the Wi-Fi access point.



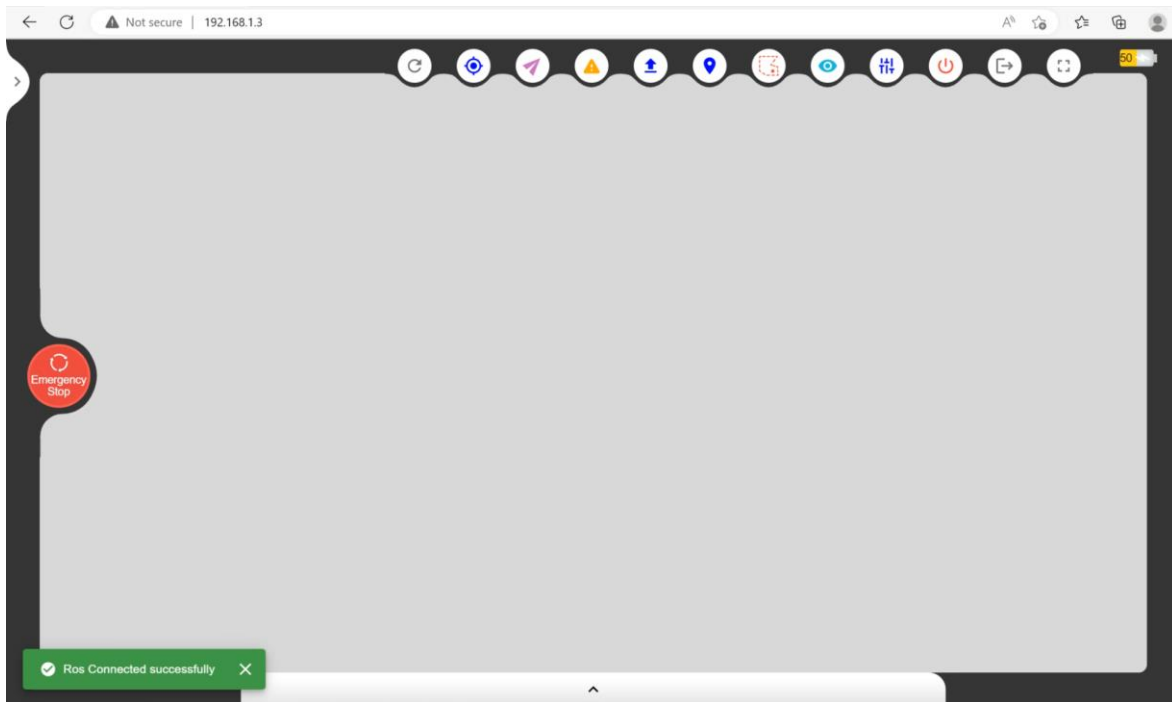
Step 2: Open your web browser and **enter the address provided to access the ANSCER Interface**



Step 3: On the tablet that ANSCER has provided to you or on your own personal tablet/PC, log in to the **ANS CER Interface** with the **username and password** that has been **provided to you**.



Step 4: You are now **signed in**. Now **switch the robot to Manual Mode** and drive the robot down the ramp created with the box panels. Steps are as described in the following Section 4.6



4.6 Operating the AMR in Manual Mode

The following instructions are for driving the AMR in Manual Mode.

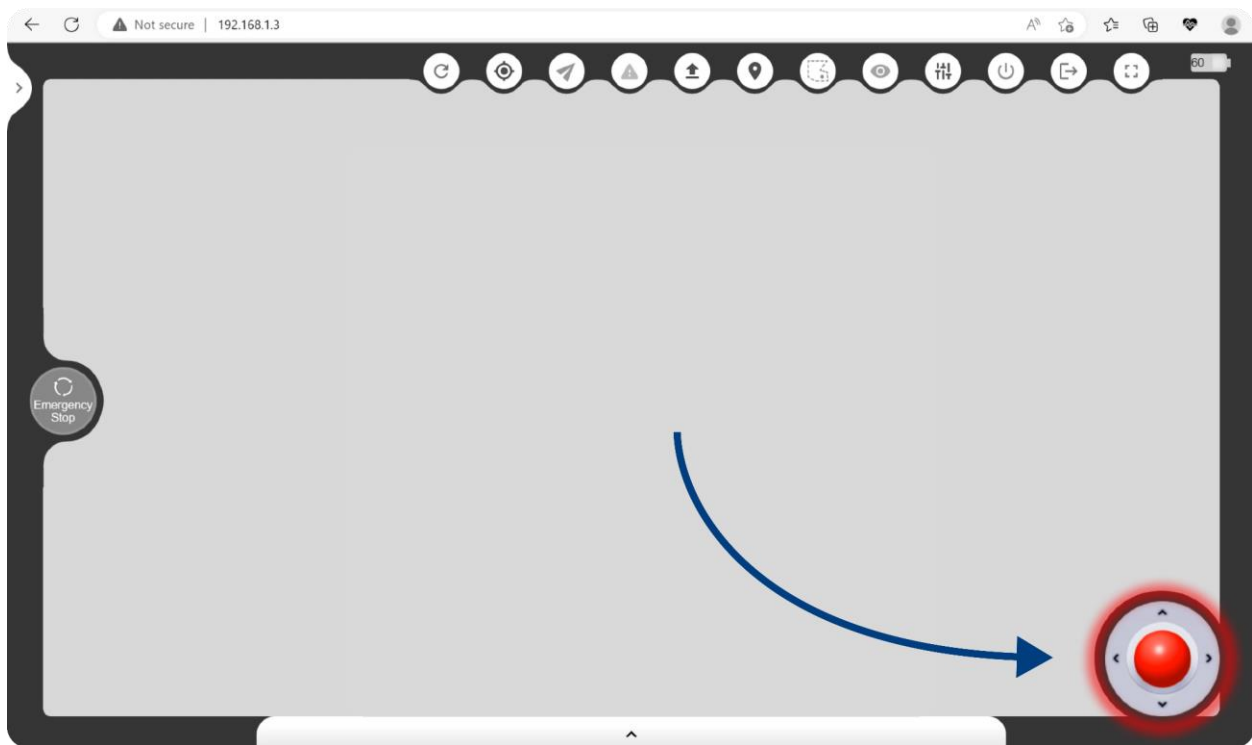
CAUTION:

When driving the AMR in Manual Mode, the operator must take extra precautions to avoid hazardous incidents like injuries and damage.

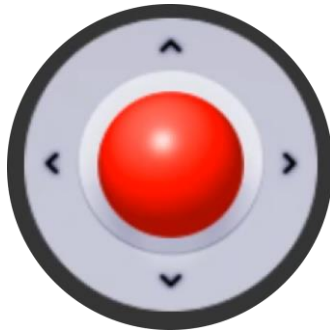
Follow the steps below to drive the AMR in Manual Mode.

Step 1: Turn the Mode Selection switch to Manual Mode.

Step 2: Identify the joystick control icon, placed at the bottom right section of the user interface.



Step 3: Use the joystick icon to control and drive the AMR manually. The LEDs will turn cyan blue, indicating that the AMR is being operated manually with the joystick.



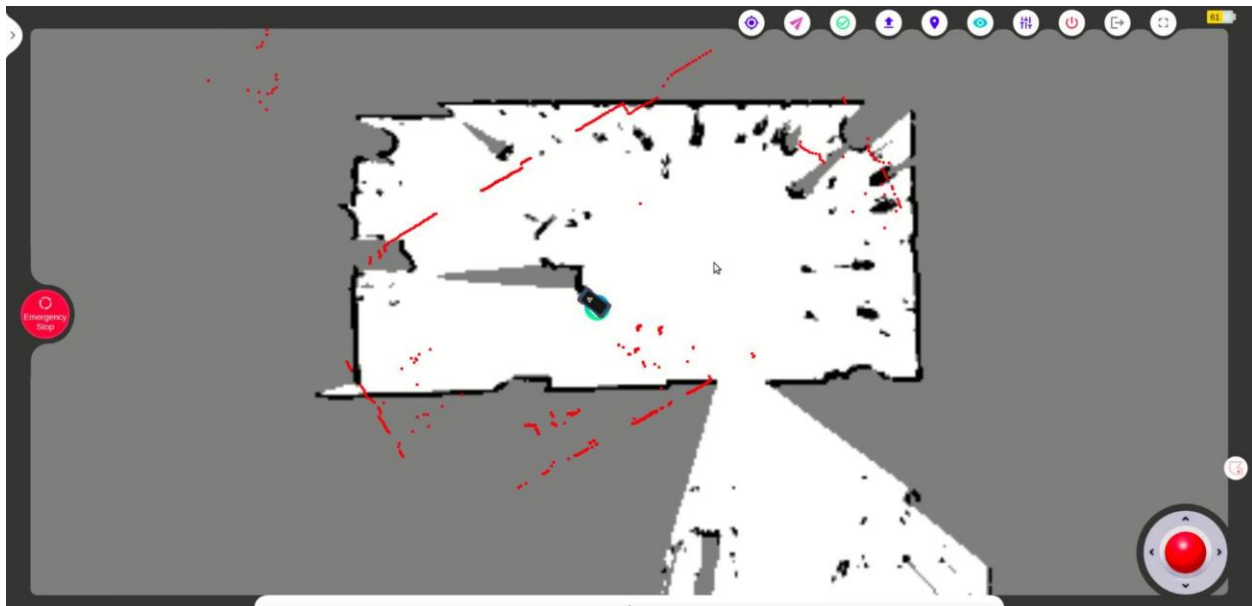
4.7 Robot Initialization

The first step after login is initialization.

Note:

As you can see in the below image, the **Lidar scan data (indicated by the red box outline)** and **the map are not aligned**. This indicates that the robot is delocalized.

We need to initialize the robot for it to operate autonomously.



Step 1: Go to the Initialize icon and select the “Manual Initialize” option.



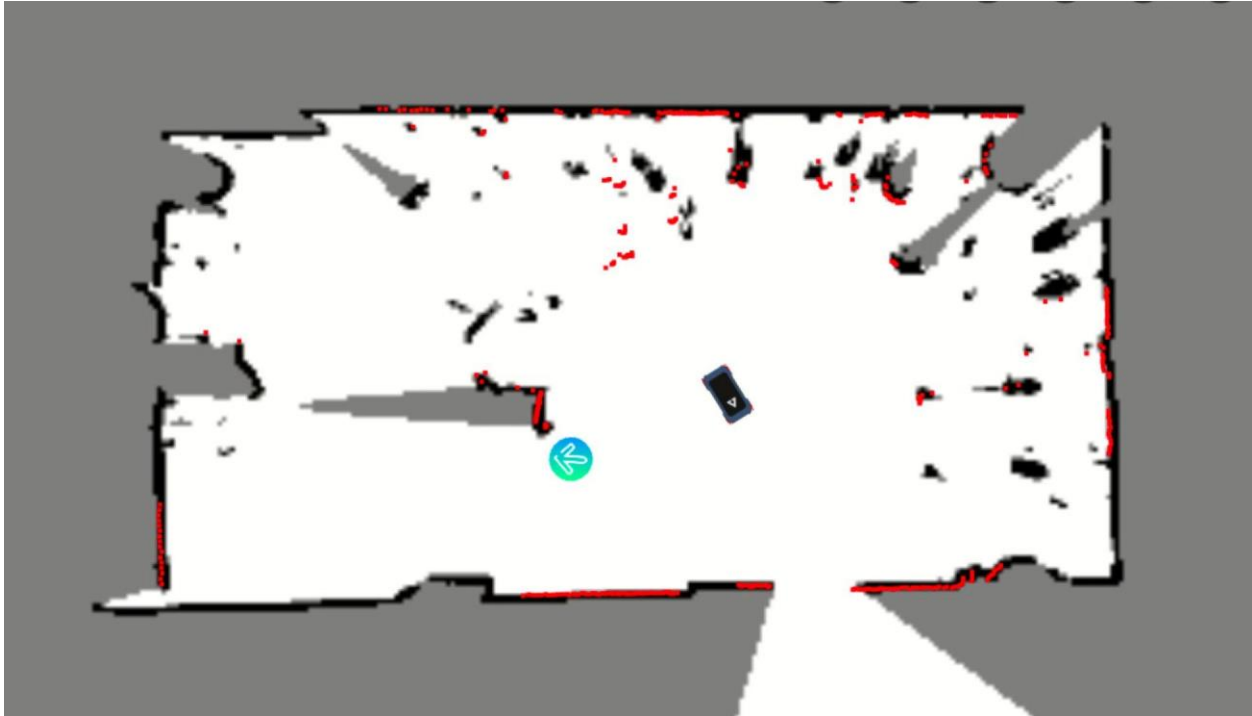
Step 2: Select the position of the robot according to the current environment and adjust the direction of the same as shown in the figure below.



Note:

If there is still a deviation between the laser scan and the existing map, select “**Auto Initialize**” from the Initialize icon.

After Initialization is complete, the Laser Scan will correctly align with the existing map. The figure below shows the same.

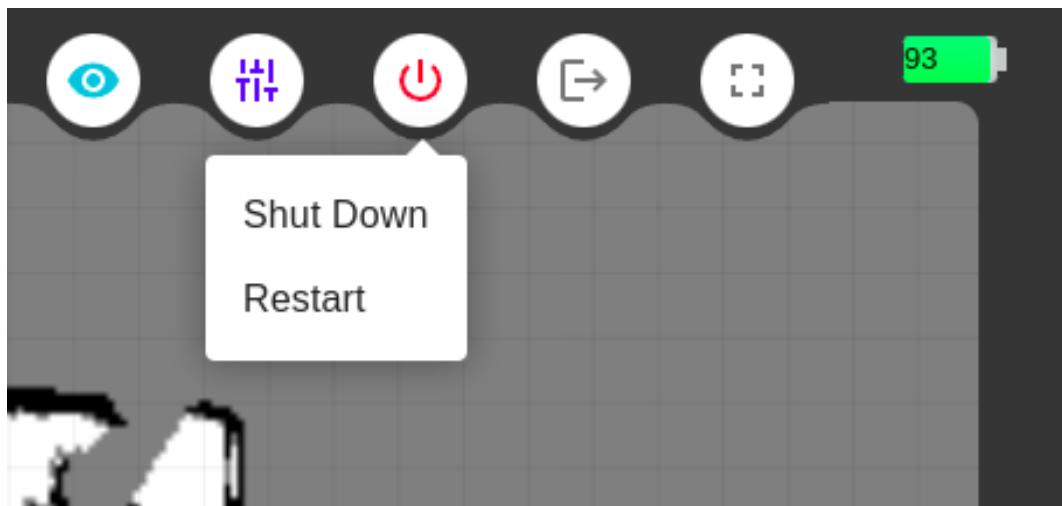


4.8 Powering Off the Robot

To power off the AMR, please follow the below steps.

Step 1: Please ensure that the AMR is stationary and not moving or in the middle of performing a task.

Step 2: Press the “Shut Down” option under the power off button that is present on the interface to switch off the connection of the tablet/PC and the AMR.



Step 3: Turn the key to the left (anticlockwise) to switch off the AMR. Once the AMR is powered off, the cyan blue lights will go off.



5. Battery and Charging

A lithium battery powers the AR100. It can be charged with an AR cable charger or a 48V AR docking station.

Please note: The AR docking station and cable charger are not available in the box and must be purchased separately for truly autonomous charging. Please contact us to know more.

5.1 Charging AR100

The charging procedure for AR100 is discussed in the following section.

There are two ways of charging the robot:

1. Auto Charging using a Docking Station
2. Manual Charging

5.1.1 Auto Charging using a Docking Station

With Auto Charging, the robot automatically goes to the docking station if the charge goes below a defined level.

5.1.2 Manual Charging

Step 1: Access the right-side panel containing the charging socket.

Step 2: Plug the cable charger in the charging point in the AMR.

Step 3: The indicator lights will turn orange to indicate the charging status of AR100.

Please note: Use only the original AR cable charger for best performance.

5.2 Disconnecting the Battery of AR100

Whenever the robot is to be transported, undergo maintenance, or stored for a long time, the battery should always be disconnected.

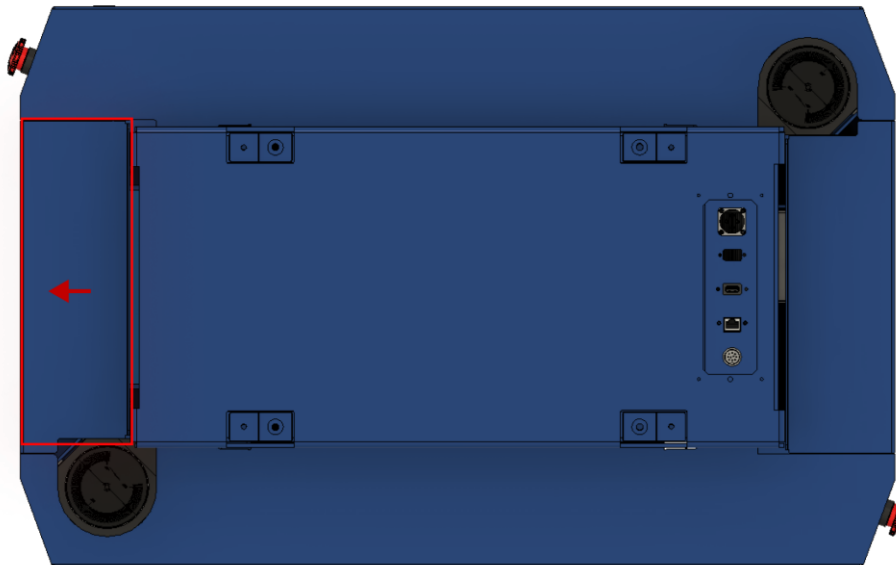
Please follow the steps below to disconnect the battery.

Step 1: Turn off the robot as described above ([Refer to Section 4.8](#))

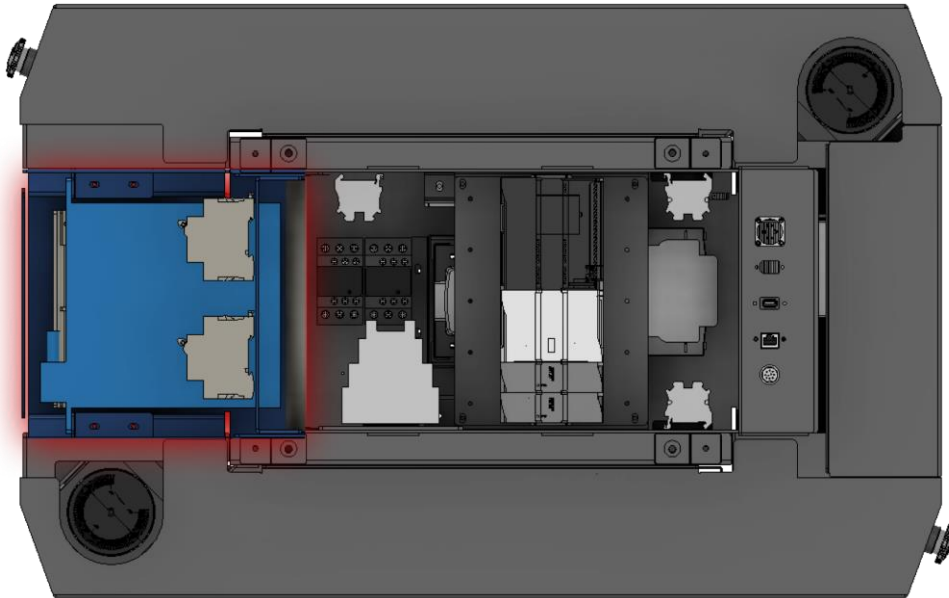
Step 2: Turn off the MCB as indicated in the image below.



Step 3: Remove the back panel to access the battery (Refer to the Service Manual for details)



Step 4: Disconnect the wire connected to the battery.



5.3 Battery Storage

The battery should be stored in an area at room temperature with a non-condensing relative air humidity temperature; humidity below or above the specifications will shorten the lifespan of the battery. The battery should not be exposed to or submerged in any liquid. Charge the battery before storage to preserve the service life of the battery.



6. Navigation and Control

The navigation and control system are used to drive the robot to its waypoints while helping it detect and avoid static and dynamic obstacles. This section explains the main components involved in the procedure.

6.1 System Overview

The user provides a map of the work environment and various waypoints for the robot. The navigation and control system ensures the AMR stays on track and reaches its goal(s).

The main processes involved here are as follows:

Global Planner

The global planner initiates the navigation process by helping the robot determine the best possible route from the starting point to its goal. The global planner ensures there are minimal obstacles in the robot's way.

Local Planner

The local planner ensures the AMR is detecting and avoiding obstacles while navigating along the path determined by the global planner.

Obstacle Detection and Avoidance

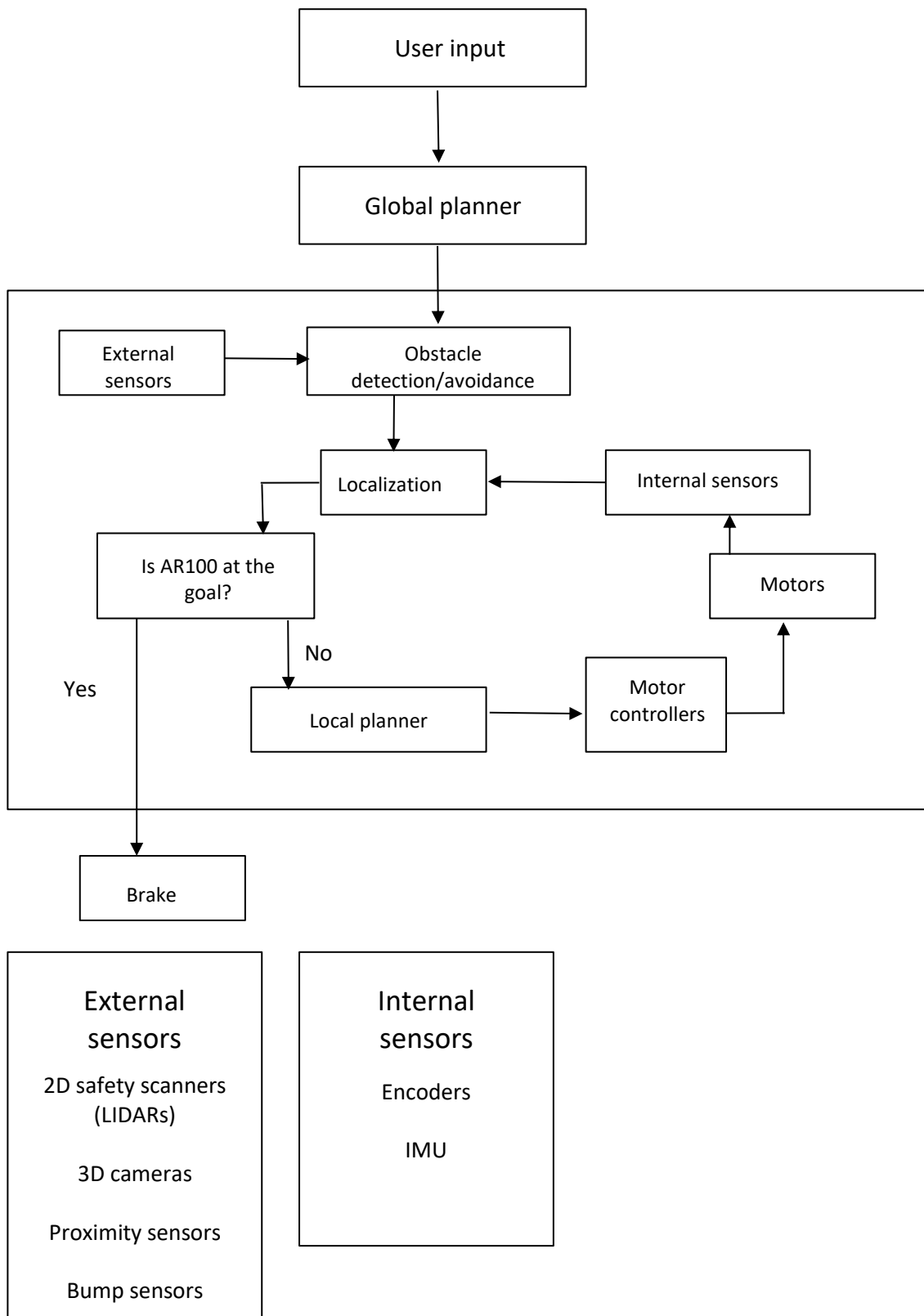
The 2D safety scanners, 3D cameras, and proximity and bump sensors help detect and avoid obstacles that may be present in the robot's path.

Localization

Localization takes input from the IMU (inertial measurement unit), 2D safety scanners, and motor encoders to understand the robot's current position on the map.

Motors and Motor Controllers

Motor controllers govern the amount of power required by the motors to drive the robot to its waypoints. Once the destination is reached, the brakes are released to halt the robot.



6.2 Obstacle Detection

The robot detects obstacles continuously while driving. This enables the robot to use the local planner to drive around obstacles and to determine the robot's current position on the map.

Three sensor types are responsible for detecting obstacles:

- External Sensors:
 1. 2D Safety Scanners
 2. 3D Cameras
- Internal Sensors:
 3. Motor Encoders

6.2.1 External Sensors

AR100 has 2 x 2D safety LIDAR scanners and 1 x 3D camera.

2D Safety Scanners

The two 2D safety scanners or static LIDARs are fixed diagonally opposite each other inside the AR100. Each of these scanners has a field view of 270°. They overlap to provide a total field view of 360° around the robot. These scanners are active and detect obstacles while the robot is moving.

Limitations:

1. The scanners can only detect obstacles with a height of at least 170 mm from the floor.
2. They cannot detect reflective and transparent objects well.
3. The safety scanners may detect false objects if exposed to strong, direct light.

Please note: If your facility has glass walls or obstacles, mark them as forbidden zones – rather than marking them as walls – to avoid disrupting the navigation system.

3D Cameras

AR100 has one 3D camera at the front. The camera can detect objects with the following specifications:

- The object is at least **280 mm** away from the robot and is at least **50 mm** in height.
- The object is at an angle of **+/-44 degrees** horizontally.

Please note: The camera is used for navigation only. It is not a part of the robot's safety system.

Limitations:

1. In contrast to the 2D safety scanners, the camera does not have a field view of 360°. It can only detect obstacles in front of the robot.
2. It does not detect holes, transparent or reflective obstacles, and descending flights of stairs.
3. When exposed to bright, direct light such as sunlight, it can hinder the camera's capabilities and cause issues in object detection.

6.2.2 Internal Sensors

Motor Encoders

The motor encoders determine the speed and distance the robot has traveled from its starting position over a period. The data received from these sensors helps provide more accurate information.

Please note: An incorrect gear ratio will lead to miscalculations based on the data provided by the motor encoders.

6.3 Localization

The localization process helps the robot figure out where it is on its map.

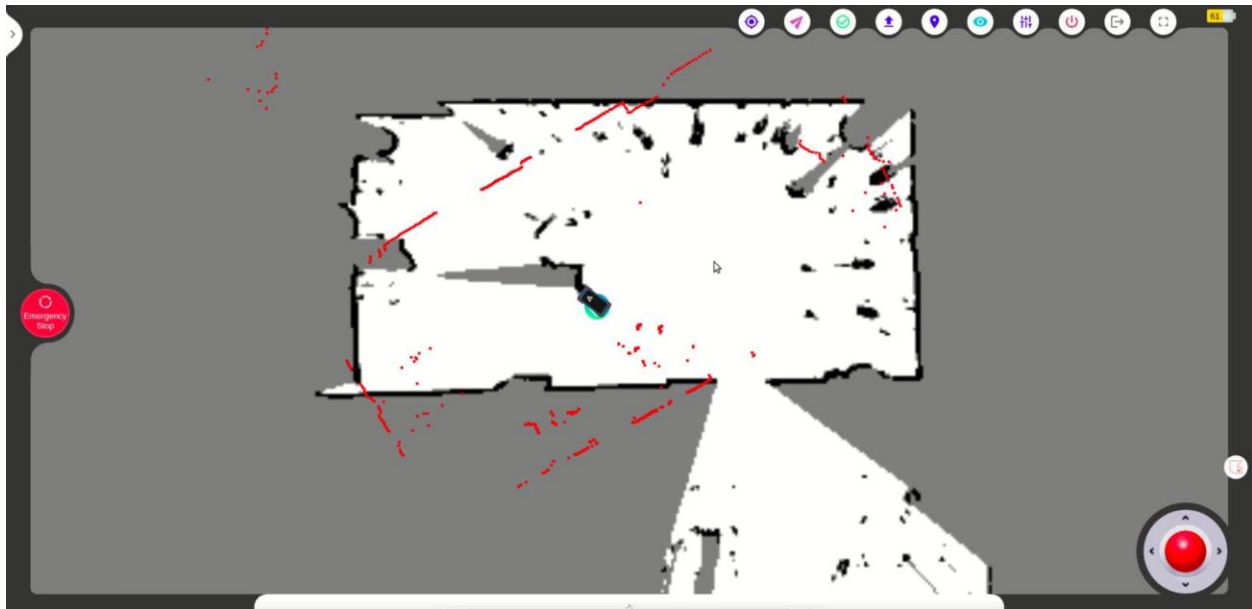
The robot has three inputs it uses to determine its location:

- Its initial position, which serves as a reference point.
- Data from its IMU and encoder, which tells it how far and fast it has traveled from its starting point.
- Data from its laser scanner, which helps it identify possible positions by comparing it with nearby walls on the map.

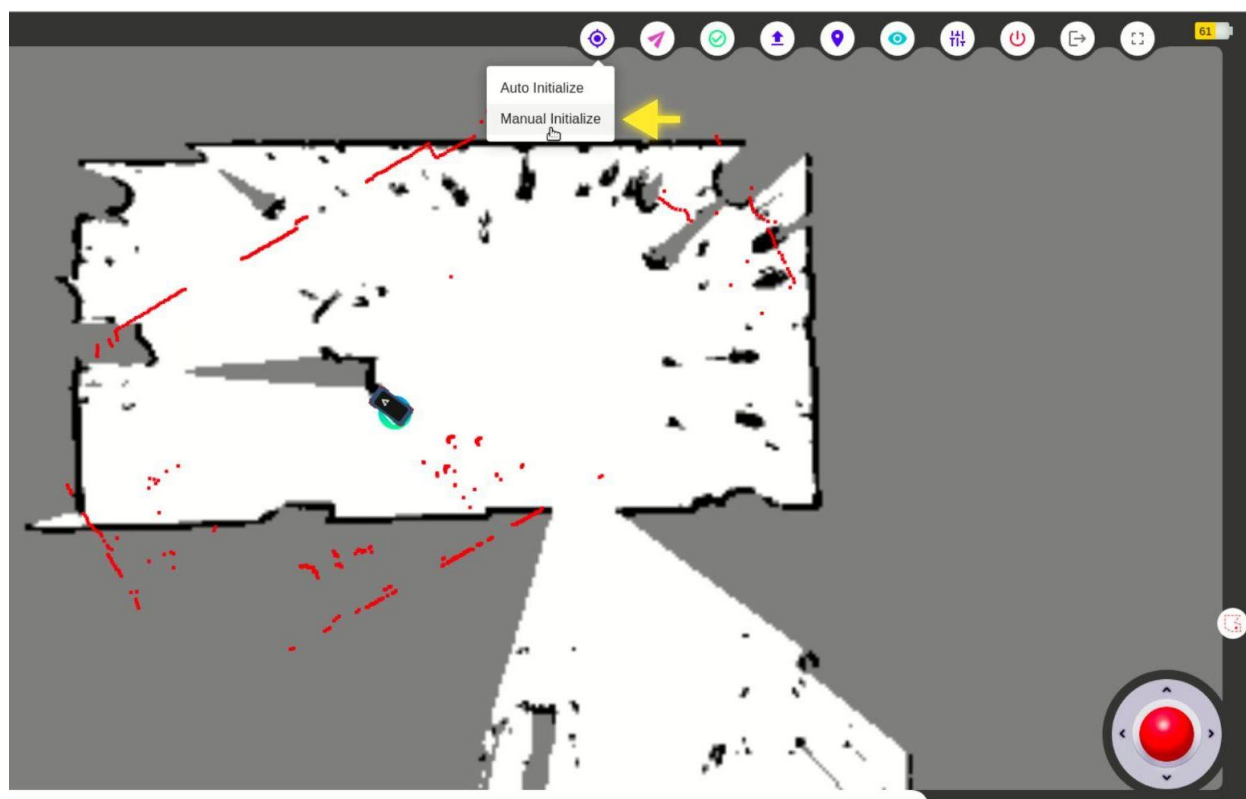
The purpose of the localization process is to enable the robot to establish its current location on its map.

Steps for Localization

The first step after login is initialization.



Step 1: Go to the Initialize icon and select the “Manual Initialize” option.



Step 2: Select the position of the robot according to the current environment and adjust the direction of the same as shown in the figure below.



Note: If there is still a deviation between the laser scan and the existing map, select “Auto Initialize” from the Initialize icon.

After Initialization is complete, the Laser Scan will correctly align with the existing map. The

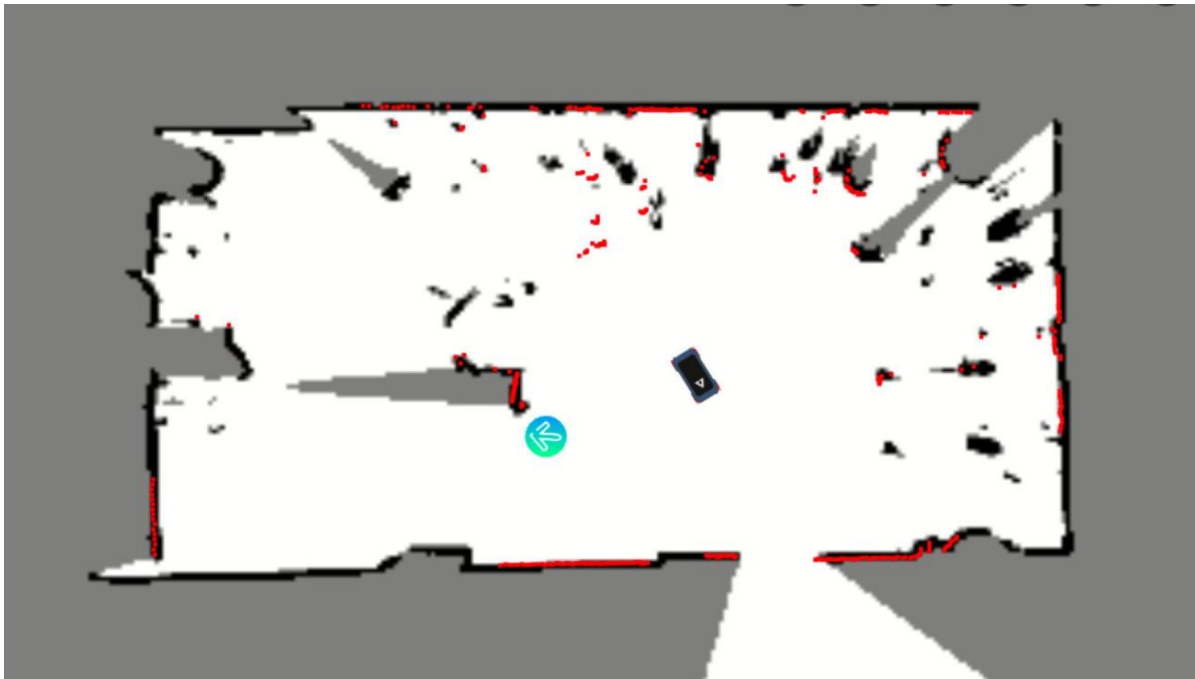


figure below shows the same.

6.4 Motors and Motor Controllers

The AR100 can moderate its speed and change its direction based on its environment. This is made possible by regulating the power sent to each motor depending on the feedback received from the sensory unit.

6.5 Brakes

The robot can gauge its real approximate position with respect to its destination on the global planner. It has a dynamic brake function that is engaged when it reaches its goal.

7. Safety System

The AR100 is designed to eliminate collision risks and effectively navigate to its waypoints without causing damage or injury.

7.1 System Overview

The PLC inside the robot controls its safety system. It is responsible for regulating inputs and outputs from operations related to safety.

Types of Stops

The AR100 has three kinds of stops. The first three types are regulated by the PLC. They are:

1. Protective Stop
2. Emergency Stop
3. Operational Stop

Protective Stop

The robot enters the Protective Stop mode when it encounters an obstacle. In this mode, the safety contactors inside the robot are switched to stop the flow of power to all its moving components. The safety contactors emit audible clicking sounds when the process initiates.

Processes During Protective Stop

The following processes are initiated when the Protective Stop is engaged:

- Motor contactors are turned off.
- Safety stop is activated.
- Motor brake is activated.

Types of Protective Stops & How to Disengage Them

- **When obstacles are detected by the 2D safety scanners in their protective field:**
Upon removing the obstacle, the robot restarts its operations after two seconds.
- **When the startup process is complete:**
Once the startup process is complete, press the Confirmation button to resume usual operations.

- **While switching between Manual and Autonomous modes:**

The robot engages Protective Stop when the operating modes are changed. It can be disengaged by pressing the Confirmation button.

Emergency Stop

The robot enters the Emergency Stop mode when the Emergency button on the AMR or the app is pressed. In this mode, the safety contactors inside the robot are switched to stop the flow of power to all its moving components. When pressed safety contactors activate with audible click sounds when the process initiates.

Once the Emergency Mode is engaged, the LED indicators turn red, and the robot stops moving until the Emergency button is released and the Confirmation button is pressed. It starts functioning again after a delay of three seconds.

CAUTION:

Pressing the Emergency stop buttons repeatedly can affect their operability. This, in turn, can cause safety hazards.

Operational Stop

Operational Stop is engaged when the AMR is halted using the ANSCER app. This can either be a part of a mission or when a mission is paused. All the modules continue to receive power from the power supply.

Safety Functions

The following functions cannot be modified or utilized with other applications and are embedded within the robot.

- **Obstacle Detection**

The robot stops if it detects an obstacle in its Protective Field. The size of the field depends upon the current speed of the robot. The higher the speed, the greater the size of the Protective Field.

- **Over-speeding Avoidance**

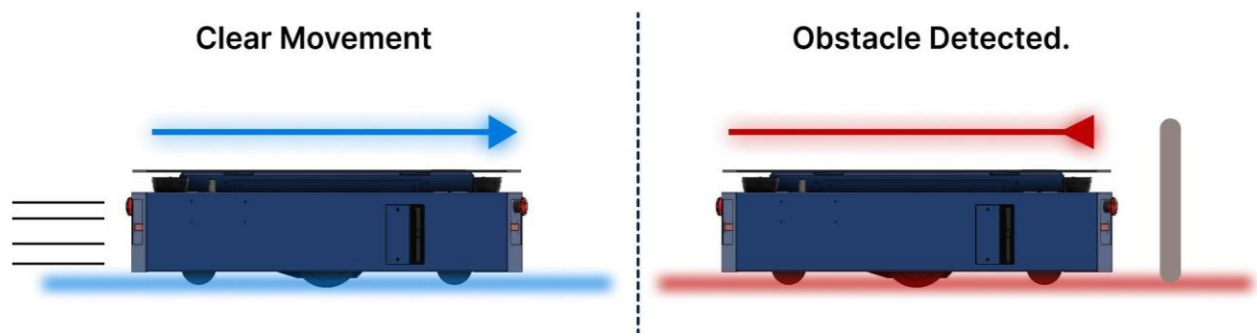
Protective stop is engaged if the safety system detects speed greater than the maximum speed of the robot.

- **Emergency Stop Buttons**

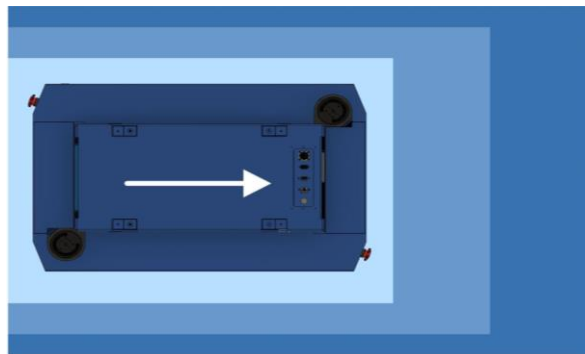
The robot enters the Emergency Stop mode when any of the two emergency buttons are pressed.

7.2 Collision Avoidance

The collision avoidance function stops the robot from crashing into people or objects by making the AMR come to a halt before it hits anything. The safety laser scanners help this function by detecting the obstacles in its way.



The safety laser scanners come equipped with two separate Protective Field sets. One field set operates when the robot is moving forward, while the other set is utilized when it is moving in reverse. These Protective Field sets are incorporated into the robot's Personnel Detection Mechanism. Each field in the sets is designed as an individually configured contour around the robot. The robot is programmed to select the correct field based on its speed. If the laser scanners detect a person or object within the active Protective Field, the robot will initiate a Protective Stop until the field is free of obstacles for a minimum of two seconds.



8. Commissioning

This section explains how to commission AR100.

Before beginning the commissioning process, the commissioning officer must ensure the following:

- There are no weights on the robot unless for a brake test, in which case the robot must carry the maximum weight that it will be required to handle regularly.
- A brake test is performed.
- The work environment is ideal.
- Forbidden or hazardous zones have been clearly marked.
- The robot installation has undergone risk assessment.
- The site has been created and configured.
- The audio and light signals have been configured in alignment with the environmental requirements.
- The ANSCER application interface has been properly configured.
- The system software has been updated to match specific requirements.

8.1 Work Environment Analysis

For the robot to function safely and as desired, its work environment must have the following specifications.

Surfaces

Slippery floors or thick carpets might hinder the operations of the robot. The floor of the work environment must also be dry for safe commute. The commissioner should ensure the robot is functioning as desired on the surface of its work environment.

Materials

Bright light or reflective and transparent objects can be hard to detect for the robot. High gloss materials can also prevent the AMR from docking successfully. The commissioner must ensure there are no such objects in the facility.

Temperature and Humidity

Temperatures and humidity outside the specifications may negatively impact the robot and its service life. The lithium-ion battery is sensitive to temperature regulations and may be damaged due to prolonged exposure to unspecified temperature conditions.

Inclines, Doorways, Sills, and Gaps

The AR100 can climb inclinations at a maximum angle of 3° with respect to a flat surface. The robot, if pushed to perform in areas that do not meet the specifications mentioned, may not be able to adequately complete its missions.

Spaces

The robot needs adequate space to dock, drive, turn, and perform its missions. The commissioner must test all the use cases of the AMR in accordance with their requirements.

Dust

The robot's performance can be affected if exposed to a dusty workspace. Dust can severely affect the robot's mechanical and sensory systems, thereby impacting its durability.

Dynamic Obstacles and Static Landmarks

For navigation, AR100 uses landmarks in its environment. If discrete markers are missing, the robot might find it difficult to navigate efficiently.

8.2 Creating and configuring Maps.

This section explains how to create a new map on the app for the AR100.

Step 1: Login to the ANSCER interface with your credentials. Ensure the robot is physically turned on.

Step 2: Connect to the robot using the dedicated Wi-Fi network.

Step 3: Once the robot is connected to the app, go to the **Menu** button present at the top of the screen and click on it.

Step 4: Click on **Maps**. Click on **Create New**. [Alternative: A pop-up will appear on the screen asking to create a map. Click on **Start Mapping**.]

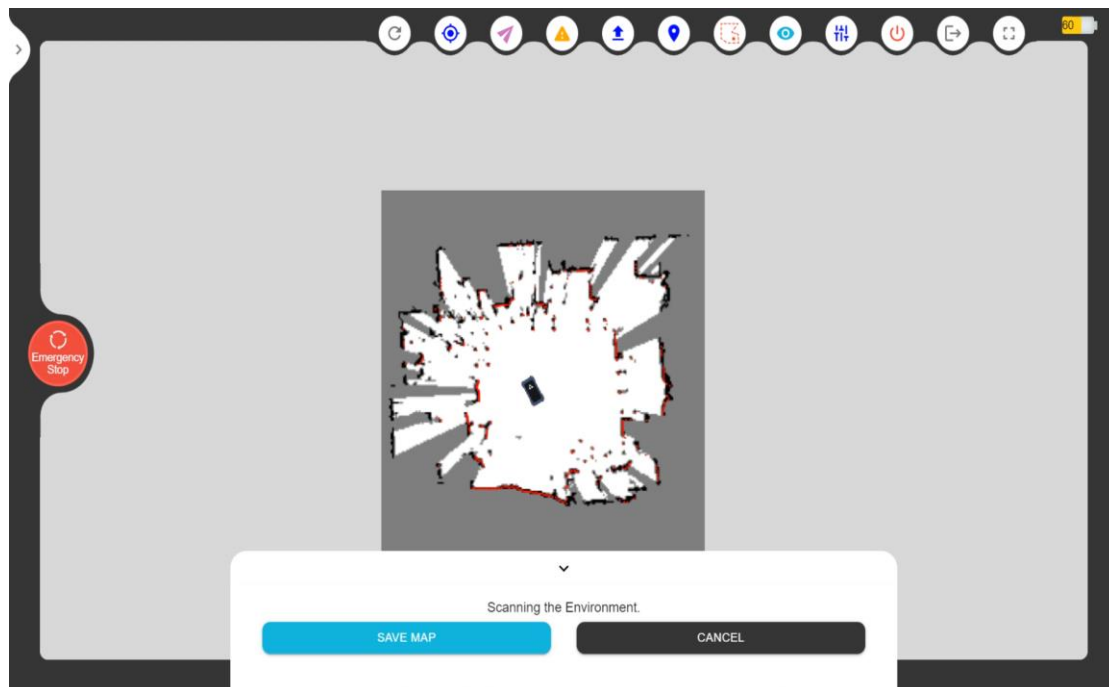
Step 5: Keep the robot at the physical position from where you want to start the map. The orientation of the robot - vertical or horizontal - should be the desired orientation of the map as visualized on the interface.

Step 6: The robot scans the environment, and the scanned output is visible on the screen. Save the map once the process is complete. [Press **Cancel** to delete this map and create a new one by following the steps above.]

Step 7: Manually localize the robot so it knows its accurate location on the map.

Under **Maps**, the active map is indicated with a green dot beside it.

Please note: All other elements in the interface, including waypoints, zones, and missions, are associated with one map, and are considered its features. Deleting a map will delete all the above. The features of one map cannot be migrated to another. New missions, waypoints, and zones must be defined for every map created.



8.3 Creating Zones on the Map

You can create zones on the map to mark specific areas. Zones are marked to trigger a specific reaction from the robot when it reaches a particular area on the map. You can mark areas where the robot is prohibited from entering as *Prohibited*; dangerous zones where the robot must slow down as *Hazardous*; and heavy traffic areas where the robot should make a sound indication as *Traffic*.

Please note: Zones on a map can have any name as long as they are meaningful for the operators.

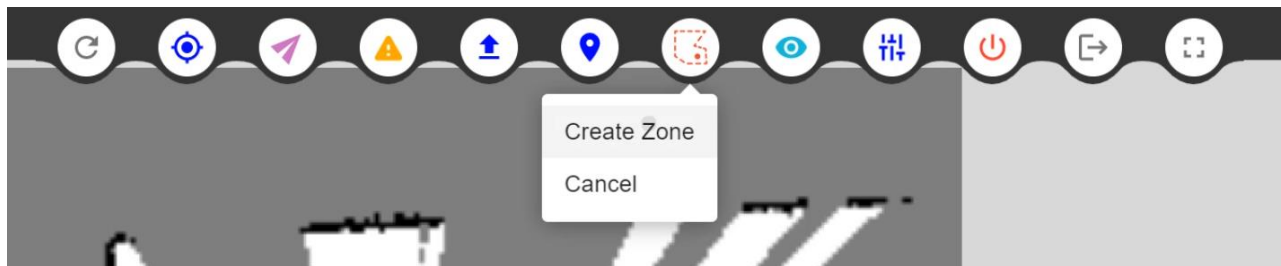
Step 1: Go to the **Create Zone** button on the dashboard.

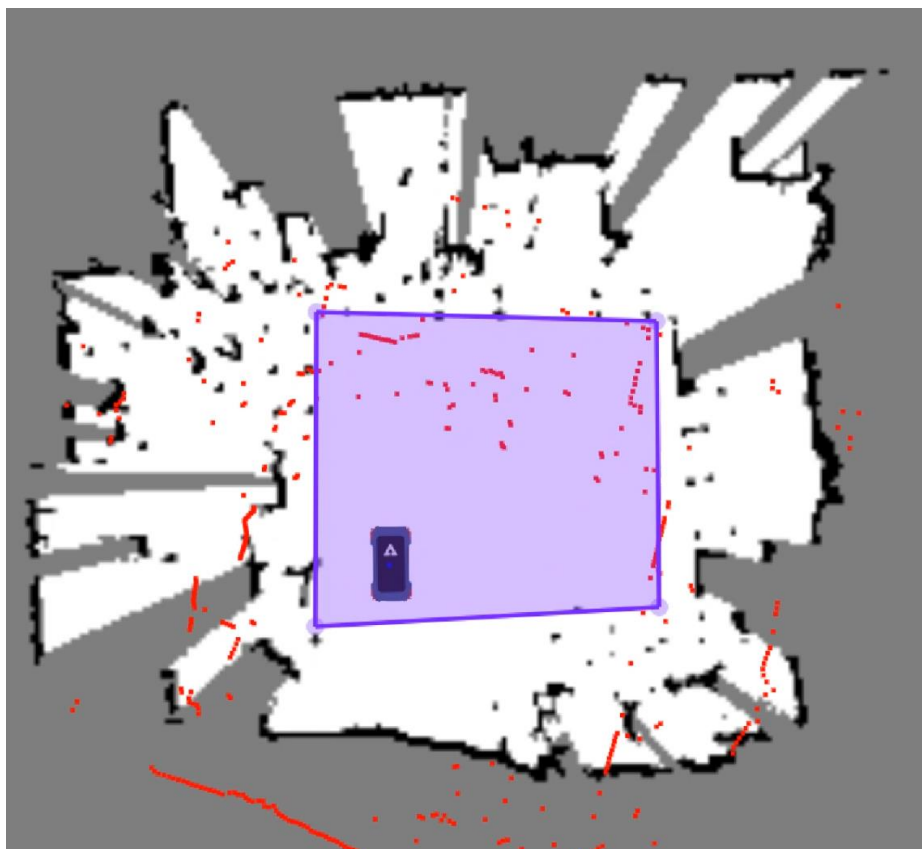
Step 2: With the cursor, mark the area on the map where the zone is.

Step 3: Set the robot behavior corresponding to the zone and give this area the desired name.

Step 4: Click on Save.

You can view and delete a zone by clicking on **Zones** under **Menu** on the dashboard.





Mark Area

Area Name

Please enter Area name

Actions in Area

☒ Restricted Zone

☐ Forbidden Zone

☐ Critical Zone

☐ Speed Zone

☐ Blink Zone

☐ Warning Zone

☐ Beep Zone



8.4 Creating Waypoints on the Map

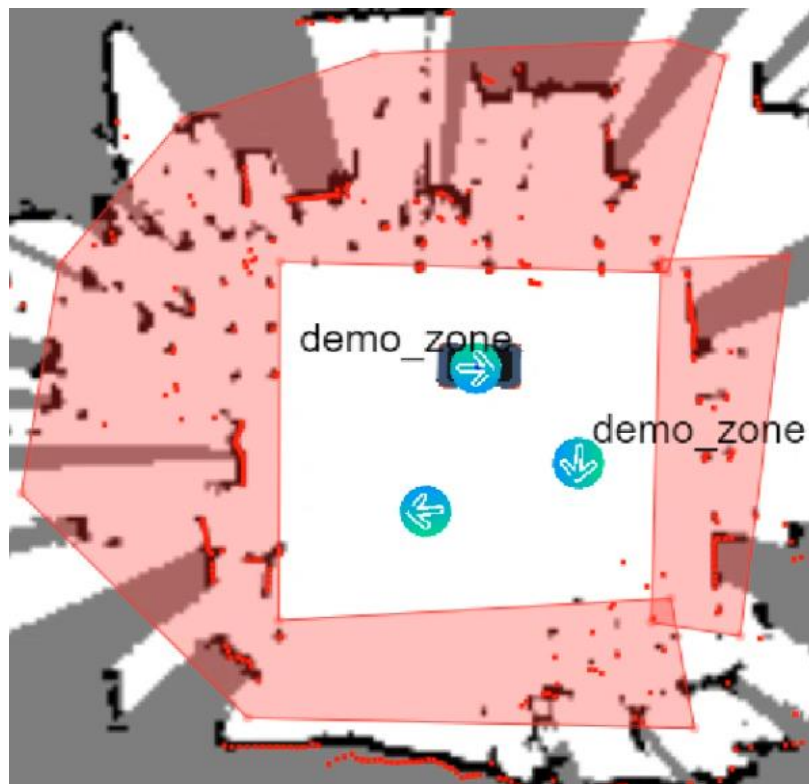
Waypoints are special coordinates on the map where a robot is supposed to reach to perform a particular task. The ANSCER app provides three types of waypoints:

Normal: The normal waypoint is a normal point on the map. For example, it could be a location where the robot must drop some loads.

Special:

Charging Station: This is the point on the map where the robot charges. Setting this waypoint helps the robot to navigate to these coordinates when it needs to be charged.

Home Position: The Home Position is a predefined point on the map that acts as the 'home' of the robot. The AR100 navigates to this waypoint when it is idle, or when it is awaiting new instructions.



8.5 Creating Missions

A mission is a collection of tasks that a robot performs at regular intervals. For example, carrying pellets from one end of a warehouse to another could be a mission. The tasks here could be picking up pellets from the starting location, dropping some of them off at one location, carrying the remaining to the next location, dropping them off at the final location, and finally returning to its home position. All of these must be defined as individual tasks under one mission. There may be multiple missions that a robot has to perform.

Follow the steps below to create and launch a mission.

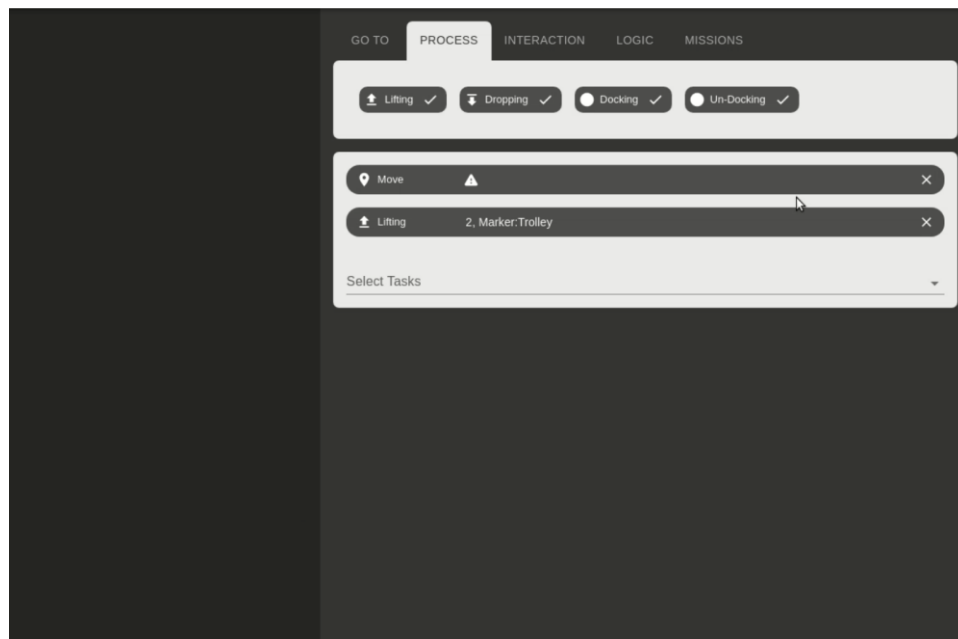
Step 1: Go to the **Menu** button on the dashboard.

Step 2: Select **Missions**.

Step 3: If you want to launch an existing mission, choose the relevant one from the missions list. If you want to create a new mission, click on **Create New**.

Step 4: Add the tasks to the new mission and launch it.

Step 5: The active mission will be indicated with a green dot beside its name.



8.6. ANSCER Application Details

The ANSCER app has twelve buttons, one emergency button, and a virtual joystick to control the robot.

Button No. (L-R)	Button	Button Operation	Button No. (L-R)	Button	Button Operation
1	Refresh	Refreshes the screen.	8	View	Enables visualization of elements on the map using a toggle button.
2	Initialize Robot	Initializes the robot physically with respect to the map. It also ensures the map is accurate with respect to the AMR's operational environment.	9	Menu	Provides access to maps, zones, waypoints, missions, and buttons.
3	Quick Goal	Helps set a quick goal like instructing the robot to reach a specific position instantly. Please note: It is not recommended to use this option repeatedly. Save a waypoint instead.	10	Power	Used to turn on and off the robot from the dashboard.
4	Status	Shows the status of the robot, battery, the robot's connectivity with the app, and localization.	11	Instant Logout	Instantly logs out of the interface.
5	Lift Up	Facilitates manual lifting of the robot.	12	Fullscreen	Helps view the app in full screen.
6	Waypoints	Sets new waypoints.		Emergency Stop	Instantly stops the robot during an emergency.
7	Create Zone	Creates specific zones on the map.		Joystick	Enables controlling the robot in manual mode. Please note: No autonomous robot functions work when it is being operated in the autonomous mode.

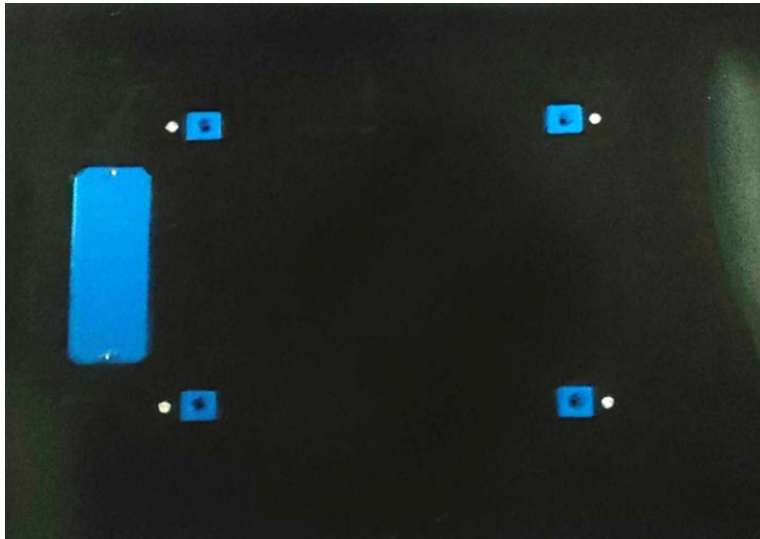
9. Applications

Add-ons can be attached to the AR100 for extended use. Operating guides explaining the attachment and usage procedures are provided with the purchase of every add-on.

Contact ANSCER Robotics for more details.

9.1 Mounting Attachments

The AR100 has four anchor points on the top plate to mount the add-ons. Refer to the image given below. The tightening torque is 72 Nm. These anchor points can also be used to fix M8 eye bolts during transportation.



CAUTION:

If the payload specifications are not followed, the AR100 may be damaged and may even cause injury to others.

10. Maintenance

This section describes the maintenance routines that are to be followed to preserve the robot's longevity.

Please note: Maintenance is the responsibility of the operator. These guidelines are relative to the frequency of robot usage and operations.

Tip: For maximum efficiency, it is recommended to maintain a maintenance calendar, so the relevant personnel is always aware of any upcoming routines.

NOTICE:

Use only original and approved spare parts for all replacements. Get in touch with ANSCER Robotics for a list of all components and replacement guides.

ANSCER Robotics is not liable for any damage caused to the AR100 due to the usage of unauthorized spare parts. The company will not be held responsible for any impairment caused to the robot, any person(s), or to the environment and facility due to the use of unapproved components in the AR100.

10.1 Weekly Checks

Carry out the following tasks once a week to ensure the robot's operational longevity.

Table 14.1a: Weekly maintenance checks

Component	Maintenance Guidelines
Top plate	Clean from the outside with a damp cloth. Do not use wet cloth, water, or compressed air.
LIDARs	Clean gently with microfiber cloth.
Castor wheels (four small wheels)	Clean with a damp cloth to remove dust and dirt particles. Remove if anything is entangled with the wheels.
Drive wheels (two center wheels)	Clean with a damp cloth to remove dust and dirt particles. Remove if anything is entangled with the wheels.

LED indicators	Clean carefully with a soft dry cloth. The cloth must not be wet or damp.
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10.2 Regular Checks and Replacements

Before performing any replacement procedures, shut down the robot and disconnect the battery.

Component	Maintenance	Maintenance Interval
Top plate	Visually check if the top plate is mounted correctly on top of the robot. Check if all connections are accessible.	Bi-annually. Replace when required.
Front, back, and side panels	Check if they are mounted correctly and are not touching the wheels.	Monthly. Replace when required.
LED indicators	Check if they are functioning correctly, including warnings.	Monthly. Replace when required.
Castor wheels	Check for wear and tear. Tighten bearings if needed.	Bi-annually. Replace when required.
Drive wheels	Check for wear and tear. Tighten bearings if needed.	Bi-annually. Replace when required. Please note: Calibrate the robot after replacing the drive wheels.
2D safety scanners	Check for dust particle contamination on the body.	Monthly. Replace when required.
Control panel	Check if all buttons are operable.	Once in 3-4 months.
Emergency buttons	Check if the Emergency buttons work. Push them to check if the reset button lights up.	Once in 3-4 months.
Charging pads	Check for dirt and dust on the charging pad.	Monthly. Replace when required.
3D cameras	Check for dust particle contamination.	Monthly. Replace when

		required.
Manual brake release switch	Release the brakes and nudge the AMR forward.	Bi-annually. Replace when required.
Nameplate and safety stickers	The nameplate, identification labels, and safety stickers must be clearly visible.	Bi-annually. Replace when required.

CAUTION:

If you suspect the robot has been damaged, perform thorough inspections to avoid injury to personnel or damage to property.

10.3 Battery Maintenance

The battery does not require maintenance but might require cleaning if it gets dirty. It must be cleaned with a soft, dry cloth only after disconnecting it from the power source.

WARNING

Do not use abrasives and solvents to clean the battery.